

Mayank Kumar

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EDUCATION

University of Central Florida

Ph.D. in Computer Science

Orlando, FL

Aug. 2023 – Present

University of Central Florida

M.S. in Computer Science

Orlando, FL

Aug. 2023 – May. 2026

Indian Institute of Technology

B.Tech. in Electrical Engineering

Patna, India

Aug. 2019 – Jun. 2023

RESEARCH AREA

My research interests include:

- **Code Generation for Secure AI Systems:** Benchmarking and improving agentic code generation for specialized ML and cryptographic workloads, with emphasis on correctness, security, and practical deployment under fully homomorphic encryption. Representative work: [ICLR'26].
- **Synthetic Data Generation:** Designing privacy-preserving synthetic data generation pipelines that maintain utility while enforcing strong confidentiality guarantees. Representative work: [ICML '26].
- **Efficient Training & Inference for Large Models:** Building systems and algorithms that reduce the compute, communication, and deployment overhead of training and inference for large-scale ML models, including privacy-preserving settings. Representative works: [ASPLOS'26], [NeurIPS'25].
- **Privacy-Preserving ML Systems:** Developing encrypted training and inference stacks plus hardware–software co-design techniques for practical, secure ML at scale. Representative works: [ICLR'26], [ASPLOS'26], [ICML '26].

RESEARCH EXPERIENCE

University of Central Florida

Graduate Research Assistant, advised by Dr. Qian Lou

Orlando, FL

Aug. 2024 – Present

- Conduct research on code generation for secure AI systems, privacy-preserving synthetic data generation, and efficient training and inference pipelines for large models under cryptographic constraints.
- Build practical systems for fully homomorphic encryption (FHE), spanning benchmark design, accelerator-aware optimizations, and communication-efficient learning for deployment-ready private ML.

WORK EXPERIENCE

University of Central Florida

Graduate Teaching Assistant

Orlando, FL

Aug. 2024 – Present

Leading labs, grading homework, and designing projects for CAP5610 Machine Learning, COP3502C Computer Science 2, and CAP4611 Algorithms for ML

Samsung Research Institute

Software Development Intern

Bangalore, India

May. 2022 – Aug. 2022

Feasibility study of using **PCA** and **ICA** for Physical Layer processing to evaluate the usage of **ML/AI for 3GPP 5G Advanced (Release 18)**.

HONORS AND AWARDS

Recipient of ICLR Student Travel Award from the ICLR conference organizers .	2026
Recipient of Presentation Fellowship from the University of Central Florida .	2026
Recipient of ORCGS Doctoral Fellowship from the University of Central Florida .	2023
Regional Mathematical Olympiad 2017 (Top 30) from Gujarat State organized by HBCSE .	2017
Ranked 8804 th in JEE-Advanced 2019 among over 150 thousand candidates.	2019

PUBLICATIONS

- [1] **Mayank Kumar**, Jiaqi Xue, Mengxin Zheng and Qian Lou. FHE-Coder: Benchmarking Secure Agentic Code Generation for Fully Homomorphic Encryption *The Fourteenth International Conference on Learning Representations, ICLR 2026*
- [4] **Mayank Kumar**, Qian Lou, Paulo L. Barreto, Martine De Cock and Sikha Pentiyala. FHAIM: Fully Homomorphic AIM for Private Synthetic Data Generation *Forty-third International Conference on Machine Learning, ICML 2026*
- [3] Fan Li, **Mayank Kumar**, Ruizhi Zhu, Mengxin Zheng, Qian Lou and Xin Xin. ReliaFHE: Resilient Design for Fully Homomorphic Encryption Accelerators *International Conference on Architectural Support for Programming Languages and Operating Systems , ASPLOS 2026*
- [2] Jiaqi Xue, **Mayank Kumar**, Yuzhang Shang, Shangqian Gao, Mengxin Zheng, Xiaoqian Jiang and Qian Lou. DictPFL: Efficient and Private Federated Learning on Encrypted Gradients. *Thirty-ninth Conference on Neural Information Processing Systems, NeurIPS 2025*